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"From Rookie to Rock"

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FREE PowerBI pbix

# Profitability of a project with UDF DAX financial pack functions

Part 1: Financial explanations and how it works?



Patou Tips #60



## About data...



The dataset uses in this Patou Tips comes from the fictitious Parisian company "IceCream' Macaron," taken from my book "Business Intelligence".

| Year | Revenue   | Cost       | Investment |
|------|-----------|------------|------------|
| 2024 | 150 000 € | -50 000 €  | -200 000 € |
| 2025 | 160 000 € | -100 000 € | -30 000 €  |
| 2026 | 190 000 € | -150 000 € | -10 000 €  |
| 2027 | 230 000 € | -150 000 € | -10 000 €  |
| 2028 | 260 000 € | -150 000 € | -10 000 €  |
| 2029 | 275 000 € | -150 000 € | -10 000 €  |
| 2030 | 300 000 € | -150 000 € | -10 000 €  |
| 2031 | 330 000 € | -150 000 € | -10 000 €  |
| 2032 | 360 000 € | -150 000 € | -10 000 €  |



→ This project represents the launch of a new store for the company "IceCream' Macaron." It involves a significant initial investment and residual investments for possible improvements to the store structure.

Use case with PowerBI

"Finance Toolbox"

Profitability of a project



## Profitability of a project

### Financial explanations (1/3)



The **profitability of a project** is essential for making informed investment decisions. This assessment, which incorporates the time value of money, the **\*\*discount factor (DF)\*\***, involves using tools such as:

The **\*\*Net Present Value (NPV)\*\*** calculates the difference between the present value of future cash flows and the initial investment. A positive NPV indicates that the project generates more value than its cost, thus signaling a profitable opportunity. Conversely, a negative NPV suggests an unviable project.

The **\*\*Internal Rate of Return (IRR)\*\*** represents the **\*\*discount factor (DF)\*\*** that makes the NPV equal to zero. It allows you to compare the project's profitability to a minimum required rate of return (cost of capital). An IRR higher than this rate means the project is attractive.

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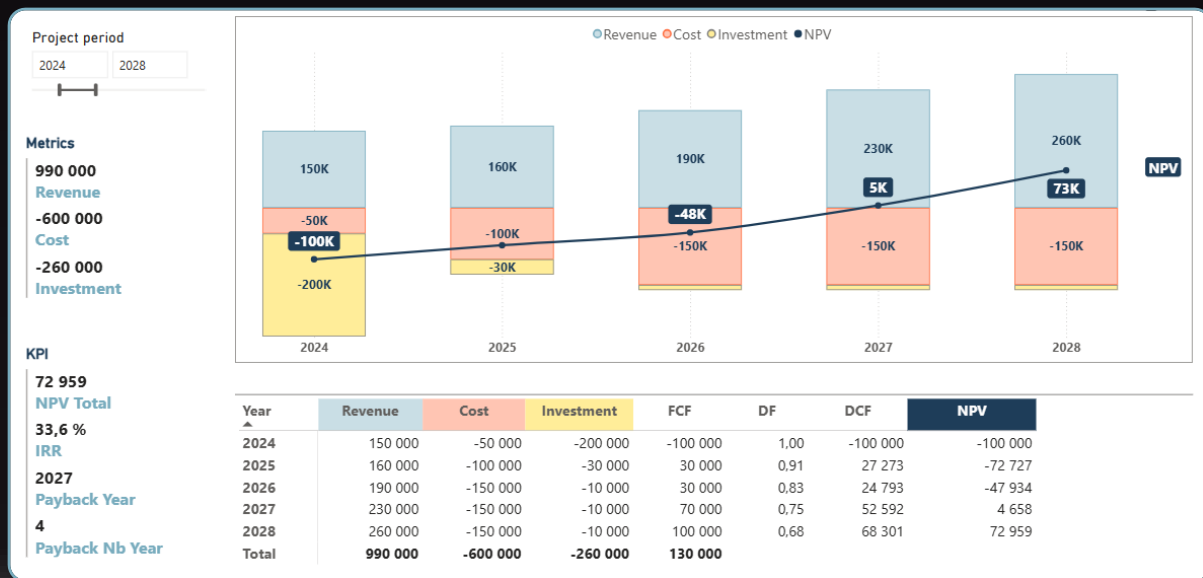
## Profitability of a project

### Financial explanations (2/3)



An indicator such as the **\*\*payback period\*\*** complements this analysis by assessing the time required to obtain a return on investment.

In summary, these tools offer a clear view of risks and opportunities, helping investors and managers to prioritize the most profitable projects, in line with their strategic objectives.



Use case with PowerBI

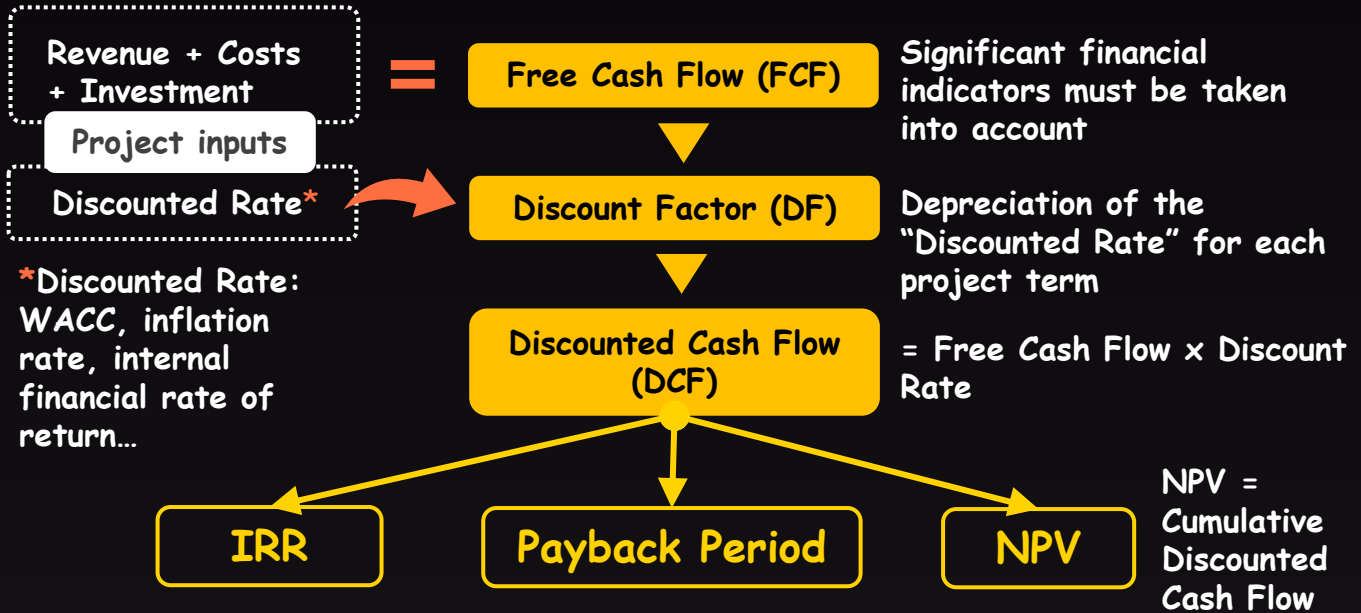
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# 0 Profitability of a project

## Financial explanations (3/3)



Profitability of a project is good when these 3 conditions (KPI) are met...

KPI



IRR > Profitability rate required



Payback period < Project Term



NPV > 0

Use case with PowerBI

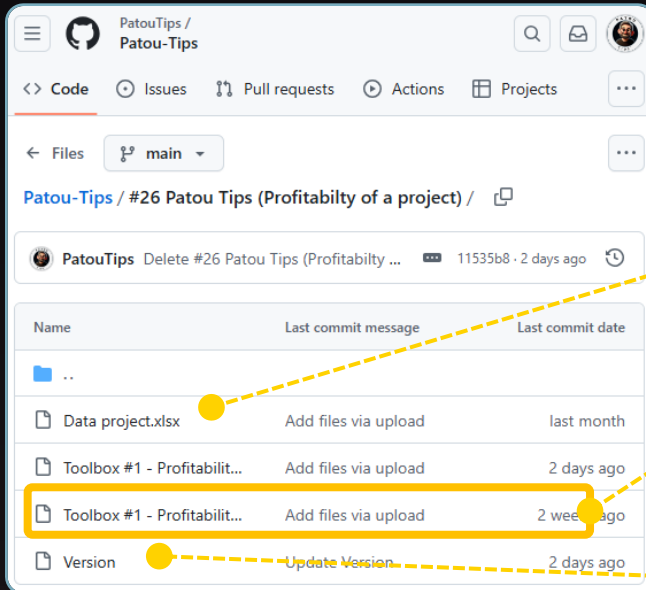
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How it works?

Profitability of a project (UDF Fiance pack)

Find all the resources  
on the Patou Tips GitHub

Data ressources (Excel file)

Toolbox #1 - Profitability of a project v2.pbix

Explanations about  
PowerBI project versions

Use case with PowerBI

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Profitability of a project

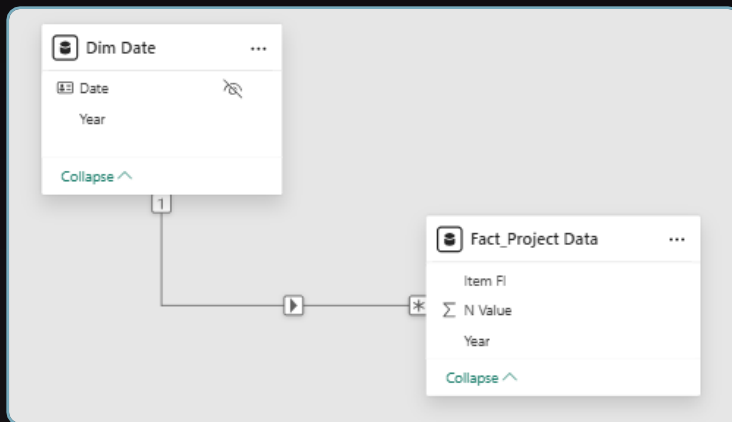
## About data...



To use the UDF of this finance pack, you need at least 3 measures :

- ✓ Revenue,
- ✓ Cost,
- ✓ Investment

Note: Cost and Investment measures be in negative values.



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**UDF** (User Defined Function) Finance Pack

# Profitability of a project

Part 1: Financial  
explanations and  
metrics





0

# Financial explanations

## Main calculation steps...



### Metrics

Revenue

Costs

Investment

Discounted Rate

Free Cash Flow (FCF)

Discount Factor (DF)

Discounted Cash Flow (DCF)

### KPI

Net Present Value  
(NPV)

Internal Rate of  
Return (IRR)

Payback Period

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# Financial explanations & DAX measures

## Free Cash Flow (FCF)



Free cash flow (FCF) is the money a company has left over after paying its operating expenses and capital expenditures.

$$\text{Free Cash Flow (FCF)} = \text{Revenue} + \text{Costs} + \text{Investment}$$

| Year | Revenue   |   | Cost       |   | Investment |   | Free Cash Flow (FCF) |
|------|-----------|---|------------|---|------------|---|----------------------|
| 2024 | 150 000 € |   | -50 000 €  |   | -200 000 € |   | -100 000 €           |
| 2025 | 160 000 € |   | -100 000 € |   | -30 000 €  |   | 30 000 €             |
| 2026 | 190 000 € |   | -150 000 € |   | -10 000 €  |   | 30 000 €             |
| 2027 | 230 000 € | + | -150 000 € | + | -10 000 €  | = | 70 000 €             |
| 2028 | 260 000 € |   | -150 000 € |   | -10 000 €  |   | 100 000 €            |
| 2029 | 275 000 € |   | -150 000 € |   | -10 000 €  |   | 115 000 €            |
| 2030 | 300 000 € |   | -150 000 € |   | -10 000 €  |   | 140 000 €            |
| 2031 | 330 000 € |   | -150 000 € |   | -10 000 €  |   | 170 000 €            |
| 2032 | 360 000 € |   | -150 000 € |   | -10 000 €  |   | 200 000 €            |

### Dax measure

1 Free Cash Flow (FCF) = CALCULATE([N])

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## Financial explanations & DAX measures

### Discount Factor (1/2)



The **Discount Factor** allows us to **evaluate the value of one euro** (or other money) according to a **required rate** of return and the **number of term**. Required rate of return is the Discounted Rate and it can be based on the WACC, the inflation rate, the financial internal rate of return...

$$\text{Discount Factor} = 1 / (1 + \text{Discounted Rate})^{(\# \text{Period} - 1)}$$

| Year | Discount rate |
|------|---------------|
| 2024 | 1,00          |
| 2025 | 0,91          |
| 2026 | 0,83          |
| 2027 | 0,75          |
| 2028 | 0,68          |
| 2029 | 0,62          |
| 2030 | 0,56          |
| 2031 | 0,51          |
| 2032 | 0,47          |

Example with a Discounted Rate of 10% and a project calculation at 2028:

$$0,68 = 1 / (1 + 10\%)^{((2028 - 2024 + 1) - 1)}$$

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# Financial explanations & DAX measures

## Discount Factor (2/2)



UDF function for Discount Factor → **DF.PatouTips**

```

FUNCTION DF.Patou.Tips =
(
    Investment_measure : expr,
    Year_date_of_calendar_table : anyref,
    Discounted_Rate : decimal
) =>

VAR First_Selected_Year =
    CALCULATE(
        FIRSTNONBLANK(Year_date_of_calendar_table, Investment_measure),
        ALLSELECTED(Year_date_of_calendar_table)
    )
VAR Last_Selected_Year =
    CALCULATE(
        MAX(Year_date_of_calendar_table),
        ALLSELECTED(Year_date_of_calendar_table)
    )
VAR Selected_Year = SELECTEDVALUE(Year_date_of_calendar_table)

VAR Number_Term = Selected_Year - First_Selected_Year + 1

VAR Discount_Factor =
    IF(
        ISINSCOPE(Year_date_of_calendar_table),
        IF(Selected_Year >= First_Selected_Year && Selected_Year <= Last_Selected_Year,
            (1/(1+Discounted_Rate)^(Number_Term-1))
        )
    )

RETURN Discount_Factor
    
```

Investment\_measure : expr,  
 Year\_date\_of\_calendar\_table : anyref,  
 Discounted\_Rate : decimal

3 informations  
 required :

- Investment
- Year in the dimension calendar table
- Discounted Rate

The first year of investment determine the beginning of the project

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## Financial explanations & DAX measures

### Discounted Cash Flow (DCF)



Discounted cash flow (DCF) is a **valuation method** that estimates the value of an investment using its **expected future cash flows**.

**Discounted Cash Flow (DCF) = Free Cash Flow x Discount Rate**

| Year | Free Cash Flow (FCF) |   | Discount rate |   | Discounted Cash Flow (DCF) |
|------|----------------------|---|---------------|---|----------------------------|
| 2024 | -100 000 €           |   | 1,00          |   | -100 000 €                 |
| 2025 | 30 000 €             |   | 0,91          |   | 27 273 €                   |
| 2026 | 30 000 €             |   | 0,83          |   | 24 793 €                   |
| 2027 | 70 000 €             | + | 0,75          | = | 52 592 €                   |
| 2028 | 100 000 €            |   | 0,68          |   | 68 301 €                   |
| 2029 | 115 000 €            |   | 0,62          |   | 71 406 €                   |
| 2030 | 140 000 €            |   | 0,56          |   | 79 026 €                   |
| 2031 | 170 000 €            |   | 0,51          |   | 87 237 €                   |
| 2032 | 200 000 €            |   | 0,47          |   | 93 301 €                   |

#### Dax measure

1 Discounted Cash Flow (DCF) = [Free Cash Flow (FCF)]\*[Discount rate]

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# Financial explanations & DAX measures NPV (Net Present Value)



When the NPV is positive, the project is adding value to by generating revenues that exceed the costs, once they are discounted.

NPV = Cumulative  
Discounted Cash Flow



NPV must be positive

| Year | Discounted Cash Flow (DCF) | NPV        |
|------|----------------------------|------------|
| 2024 | -100 000 €                 | -100 000 € |
| 2025 | 27 273 €                   | -72 727 €  |
| 2026 | 24 793 €                   | -47 934 €  |
| 2027 | 52 592 €                   | 4 658 €    |
| 2028 | 68 301 €                   | 72 959 €   |
| 2029 | 71 406 €                   | 144 365 €  |
| 2030 | 79 026 €                   | 223 392 €  |
| 2031 | 87 237 €                   | 310 629 €  |
| 2032 | 93 301 €                   | 403 930 €  |

This project starts creating value from the fourth year, means that this project must have at least 4 years as a term to be profitable

## Dax measure (2 variants)

With XNPV function

```
1 NPV DAX =
2 XNPV('Project Data',
3     [Discounted Cash Flow (DCF)],
4     'Project Data'[Year],
5     [Discounted Cash Flow (DCF)])
```

With cumulative iteration. Better to display for each year.

```
1 NPV =
2 -- NPV (Net Present Value) = Cumulative Discounted Cash flow
3 VAR Selected_Year = SELECTEDVALUE(DimDate[Year])
4
5 RETURN
6 IF(Selected_Year<=[Last Year BP],
7     CALCULATE(
8         SUMX('Project Data',[Discounted Cash Flow (DCF)]),
9         DimDate[Year]<=Selected_Year
10    )
11 )
```

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Profitability of a project

## 5

## Financial explanations & DAX measures IRR (Internal Rate of Return)



The IRR is the **discount rate** at which the net present value (NPV) of a set of **discounted cash flows is equal to zero**. On PowerBI or Excel, creating such a calculation is so complex, fortunately there is the "XIRR" function.

### Dax measure

```
1 IRR =  
2 VAR First_Term_Year = [First Year Project]  
3 VAR Last_Term_Year_Selected = [Last Year Term Selected]  
4  
5 RETURN  
6 CALCULATE(  
7     XIRR('Project Data',[Free Cash Flow (FCF)],'Project Data'[Year]),  
8     FILTER(  
9         DimDate,  
10        DimDate[Year] >= First_Term_Year && DimDate[Year] <= Last_Term_Year_Selected))
```



### IRR limits:

- The IRR assumes that the project's future cash flows are reinvested at the same rate as the IRR, which may not be realistic or achievable. ...
- Sometimes project A may have a higher IRR than project B while project B generates more cash flow (NPV).

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# Financial explanations & DAX measures

## Payback Period



The **payback period** is the amount of time it takes to recover the cost of an investment. Simply put, it is the length of time an investment reaches a **breakeven point**.

A payback period relatively short indicates a rapid return on investment, that reduces risk and improves liquidity.

**Payback Period = (Project Start Date - Date when NPV >= 0) +1**

### Dax measure



The payback period must be less than or within the "project duration"

```
1 Payback Period =
2 VAR Payback_Period_End =
3 year(CALCULATE(
4     MIN('Project Data'[Year]),
5     FILTER(
6         ADDCOLUMNS(
7             'Project Data',
8             "CumulativeCashFlow",
9             CALCULATE(
10                SUMX('Project Data',[Free Cash Flow (FCF)]*[Discount rate]),
11                FILTER(
12                    ALL('Project Data'),
13                    'Project Data'[Year] <= EARLIER('Project Data'[Year])
14                )
15            )
16        ),
17        [CumulativeCashFlow] >= 0
18    )
19 ))
20
21 RETURN Payback_Period_End-[First Year Project]+1
```

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Profitability of a project



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# Profitability of a project

Part 2: How to use



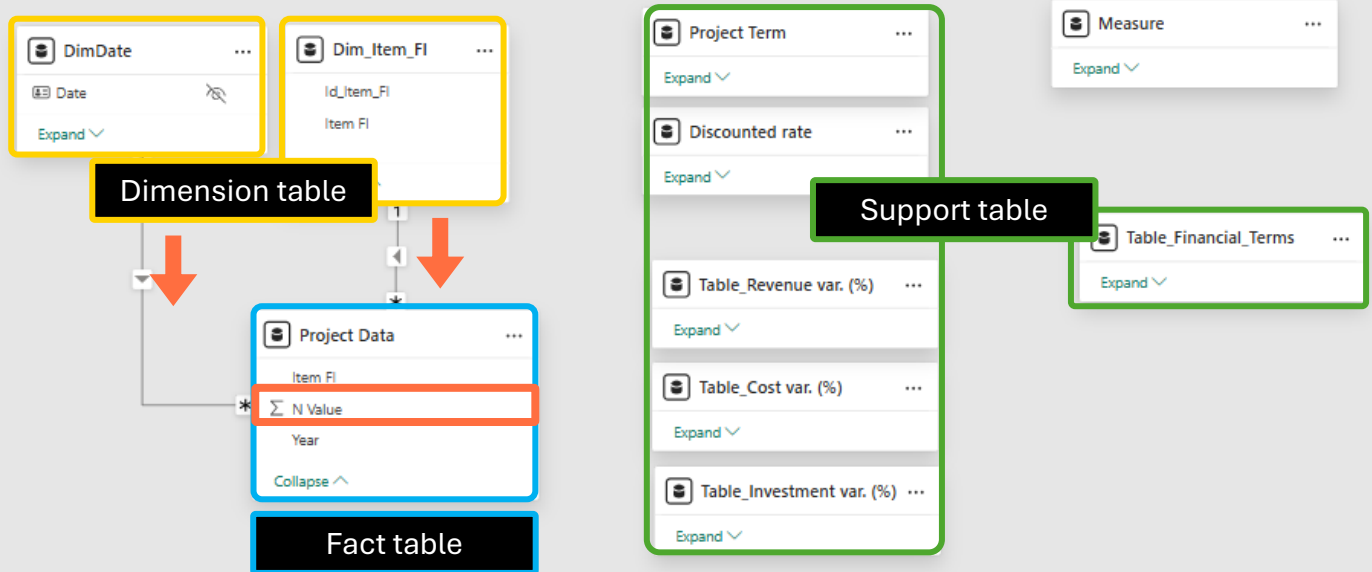


1

# Explanation of modeling Main architecture...

## Star schema

The "N value"\* is in the heart of the project!



\*See Patou Tips # 25 "The complete guide to start with PowerBI" for more information.

Use case with PowerBI

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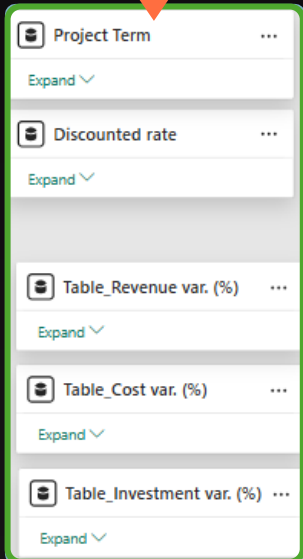
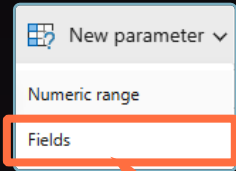
Profitability of a project



1

# Explanation of modeling Main architecture...

## About "support tables"



"Support tables" are created by the parameter functionality "fields" to adjust project inputs: project term, discounted rate, revenue, cost and investment.

Project Input

Project Term (Year)

Discounted Rate

4

10,0 %

► The chosen duration of the project extends from 2024 til 2027 (4 years) with a "Discounted Rate" of 10,00 %

Revenue var. (%)

Cost var. (%)

Investment var. (%)

0 %

Initial value

730 000,00 €

Changed value

730 000,00 €

Reset

0 %

Initial value

-450 000,00 €

Changed value

-450 000,00 €

Reset

0 %

Initial value

-250 000,00 €

Changed value

-250 000,00 €

Reset

Use case with PowerBI

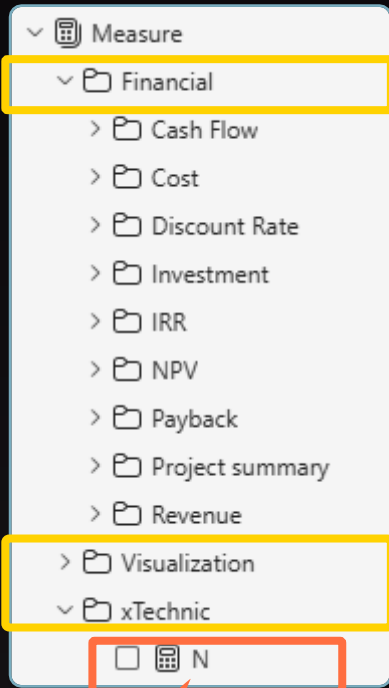
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Profitability of a project

2

## Details on DAX measures

### About measures organization



All metrics are organized into two main folders:

- Finance: This folder groups metrics by financial theme.
- Visualization: This folder allows you to add metrics to build the visualization.

A third folder "xTechnic" contains a single metric in this project, allowing to obtain the "N" metric relative to the "n value" in the fact table. This metric is the axis for all other metrics.

```
1 N = SUM('Project Data'[N Value])
```

Use case with PowerBI

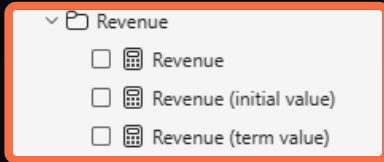
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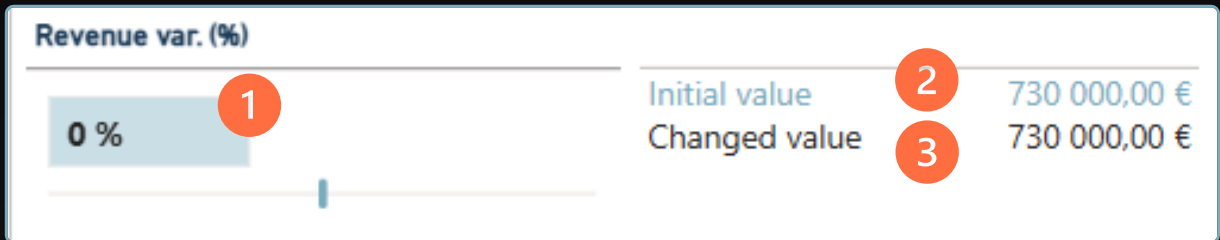
2

## Details on DAX measures Revenue (1/2)

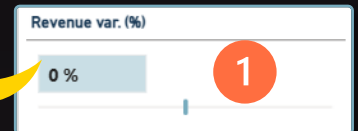


The measures of this folder are also the same for the Cost and Investment folders

Project input box



```
1 Revenue =  
2 CALCULATE(  
3 [N]*(1+'Table_Revenue var. (%)'[Revenue var. (%) Value]),  
4 FILTER('Project Data','Project Data'[Item FI]="Revenue")  
5 )
```



This metric create the "Revenue" value and takes into account the adjusted input value of the project input box. This measure will be use for the visualizations.

Use case with PowerBI

"Finance Toolbox"

Profitability of a project



2

## Details on DAX measures Revenue (2/2)

2

```
1 Revenue (initial value) =  
2 VAR First_Term_Year = [First Year Project]  
3 VAR Last_Term_Year_Selected = [Last Year Term Selected]  
4  
5 RETURN  
6 CALCULATE(  
7     [N],  
8     FILTER('Project_Data', 'Project_Data'[Item_ET]="Revenue"),  
9     DimDate[Year]>= First_Term_Year && DimDate[Year] <= Last_Term_Year_Selected)
```

|               |              |
|---------------|--------------|
| Initial value | 730 000,00 € |
|---------------|--------------|

Term of the  
project

This metric DOES NOT take into account the adjusted input value from the project input box, but calculates revenue during the term of the project. It will be useful to calculate the "initial value" of the project's revenue before any adjustments.

3

```
1 Revenue (term value) =  
2 VAR First_Term_Year = [First Year Project]  
3 VAR Last_Term_Year_Selected = [Last Year Term Selected]  
4  
5 RETURN  
6 CALCULATE([Revenue],  
7     DimDate[Year]>= First_Term_Year && DimDate[Year] <= Last_Term_Year_Selected)
```

|               |              |
|---------------|--------------|
| Changed value | 730 000,00 € |
|---------------|--------------|

Term of the  
project

This metric take into account the adjusted input value from the project input box, but calculates revenue during the term of the project. It will be useful to calculate the "changed value" of the project's revenue after adjustments.

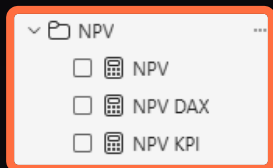
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2

## Details on DAX measures NPV (Net Present Value)



```
1 NPV DAX =
2 XNPV('Project Data',
3     [Discounted Cash Flow (DCF)],
4     'Project Data'[Year],
5     [Discounted Cash Flow (DCF)])
```

In this project, I'm not using the native "XNPV" function to calculate the NPV. I created it purely for learning purposes.

I prefer to use a cumulative DAX pattern to obtain a value for each year.

```
1 NPV =
2 -- NPV (Net Present Value) = Cumulative Discounted Cash flow
3 VAR Selected_Year = SELECTEDVALUE(DimDate[Year])
4
5 RETURN
6 IF(Selected_Year<=[Last Year BP],
7     CALCULATE(
8         SUMX('Project Data',[Discounted Cash Flow (DCF)]),
9         DimDate[Year]<=Selected_Year
10    )
11 )
```

Cumulative DAX pattern

```
1 NPV KPI =
2 VAR End_Period = [Last Year Term Selected]
3
4 RETURN
5 CALCULATE(
6     [NPV],
7     DimDate[Year]=End_Period)
```

| KPI ROI Project |                |
|-----------------|----------------|
| 4               | Payback (year) |
| 12,25 %         | IRR            |
| 4 658 €         | NPV            |

This metric is created to show the ending value at the project final term, to be displayed for the KPI visualization.

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Profitability of a project

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## Financial explanations & DAX measures IRR (Internal Rate of Return)



Creating such a calculation in PowerBI or Excel is complex, because for each project period, you need to calculate which rate gives a zero value for the NPV. Fortunately, there is the "XIRR" function.

```
1 IRR =  
2 VAR First_Term_Year = [First Year Project]  
3 VAR Last_Term_Year_Selected = [Last Year Term Selected]  
4  
5 RETURN  
6 CALCULATE(  
7     XIRR('Project Data',[Free Cash Flow (FCF)], 'Project Data'[Year]),  
8     FILTER(  
9         DimDate,  
10        DimDate[Year] >= First_Term_Year && DimDate[Year] <= Last_Term_Year_Selected))
```

Term of the project

Use case with PowerBI

"Finance Toolbox"

Profitability of a project



2

# Financial explanations & DAX measures

## Payback Period



Payback

☐ Payback Period

☐ Payback Period (Year)

```

1 Payback Period =
2 VAR Payback_Period_End =
3 year(CALCULATE(
4     MIN('Project Data'[Year]),
5     FILTER(
6         ADDCOLUMNS(
7             'Project Data',
8             "CumulativeCashFlow",
9             CALCULATE(
10                SUMX('Project Data',[Free Cash Flow (FCF)]*[Discount rate]),
11                FILTER(
12                    ALL('Project Data'),
13                    'Project Data'[Year] <= EARLIER('Project Data'[Year])
14                )
15            )
16        ),
17        [CumulativeCashFlow] >= 0
18    )
19 ))
20
21 RETURN Payback_Period_End-[First Year Project]+1
  
```

Cumulative DAX pattern

This formula calculate the first year (line 4) of the Payback Period when the value of the NPV (lines 7 to 13) fill the condition NPV is  $\geq 0$  (line 17).

```
1 Payback Period (Year) = [Payback Period]&" years"
```

This formula allow

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Profitability of a project

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# Profitability of a project

Part 3: Graphic  
Design, UI/UX, and  
Storytelling

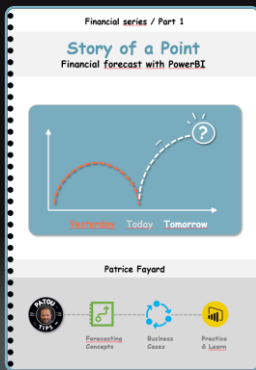


Don't forget!  
This isn't the truth, it's just my truth!



## Patou Tips

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